

Homework 1

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Problem 1. Random variables

A random variable encodes a numerical feature of an experiment, allowing us to study its distribution and moments rather than every detail of the outcome. For example, the number of heads in two coin tosses maps both HT and TH to 1. Formally, $X : \Omega \rightarrow \mathbb{R}$ is a measurable function: the mapping is fixed, but the realized value $X(\omega)$ depends on the random outcome ω . This uncertain numerical value explains the name “random variable.”

Problem 2. Probability and estimation identities

Write p for probability mass functions and f for densities. Density identities hold almost everywhere, and conditional ratios require positive denominators. Assume the densities and finite second moments used below exist.

(a) General Bayes’ rule: four cases

The joint mass/density can be factored in either order. With rows specifying the types of (X, Y) , these factorizations give

discrete, discrete	$p_{Y X}(y x) = \frac{p_{X Y}(x y)p_Y(y)}{p_X(x)},$
discrete, continuous	$f_{Y X}(y x) = \frac{p_{X Y}(x y)f_Y(y)}{p_X(x)},$
continuous, discrete	$p_{Y X}(y x) = \frac{f_{X Y}(x y)p_Y(y)}{f_X(x)},$
continuous, continuous	$f_{Y X}(y x) = \frac{f_{X Y}(x y)f_Y(y)}{f_X(x)}.$

For example, in the mixed discrete–continuous case, for every measurable B ,

$$p_X(x) \int_B f_{Y|X}(y | x) dy = \mathbb{P}(X = x, Y \in B) = \int_B p_{X|Y}(x | y) f_Y(y) dy.$$

Equality of the integrands proves the second formula; exchanging X, Y proves the third. The first and fourth follow directly by factoring $p_{X,Y}$ and $f_{X,Y}$, respectively.

(b) General law of total probability: four cases

Marginalize the joint distribution over Y (sum for discrete Y , integrate for continuous Y), then substitute its conditional factorization:

discrete, discrete	$p_X(x) = \sum_y p_{X Y}(x y)p_Y(y),$
discrete, continuous	$p_X(x) = \int_{\mathbb{R}} p_{X Y}(x y)f_Y(y) dy,$
continuous, discrete	$f_X(x) = \sum_y f_{X Y}(x y)p_Y(y),$
continuous, continuous	$f_X(x) = \int_{\mathbb{R}} f_{X Y}(x y)f_Y(y) dy.$

The first two formulas sum or integrate $\mathbb{P}(X = x, Y \in \cdot)$. For the last two, apply the same argument to $\mathbb{P}(X \in A)$ for every measurable A , interchange nonnegative integrals/sums by Tonelli’s theorem, and identify the density of X .

(c) Law of total variance

Let $m(X) = \mathbb{E}[Y | X]$. The tower property gives $\mathbb{E}[m(X)] = \mathbb{E}[Y]$, so

$$\begin{aligned}\mathbb{E}[\text{Var}(Y | X)] + \text{Var}(m(X)) &= \mathbb{E}[Y^2] - \mathbb{E}[m(X)^2] + \mathbb{E}[m(X)^2] - (\mathbb{E}[Y])^2 \\ &= \text{Var}(Y).\end{aligned}$$

(d) Linear least-squares estimate

For $\text{Var}(X) > 0$, centering yields

$$\mathbb{E}[(Y - a - bX)^2] = \text{Var}(Y) - 2b \text{Cov}(X, Y) + b^2 \text{Var}(X) + (\mathbb{E}[Y] - a - b\mathbb{E}[X])^2.$$

First choose $a = \mathbb{E}[Y] - b\mathbb{E}[X]$; the remaining strictly convex quadratic is minimized at $b = \text{Cov}(X, Y) / \text{Var}(X)$. Hence

$$L[Y | X] = \mathbb{E}[Y] + \frac{\text{Cov}(X, Y)}{\text{Var}(X)}(X - \mathbb{E}[X]).$$

If $\text{Var}(X) = 0$, every affine predictor is constant almost surely and the best predicted value is $\mathbb{E}[Y]$.

(e) Orthogonality of the MMSE estimate and its converse

For $m(X) = \mathbb{E}[Y | X]$ and any square-integrable $\phi(X)$,

$$\mathbb{E}[(Y - m(X))\phi(X)] = \mathbb{E}[\phi(X)\mathbb{E}[Y - m(X) | X]] = 0.$$

Conversely, if a square-integrable $g(X)$ satisfies the same condition for every such ϕ , subtraction gives $\mathbb{E}[(m(X) - g(X))\phi(X)] = 0$. Taking $\phi(X) = m(X) - g(X)$ yields $\mathbb{E}[(m(X) - g(X))^2] = 0$, so $g(X) = \mathbb{E}[Y | X]$ almost surely. In particular, orthogonality implies

$$\mathbb{E}[(Y - g(X))^2] = \mathbb{E}[(Y - m(X))^2] + \mathbb{E}[(m(X) - g(X))^2],$$

which also proves the MMSE property.

(f) A two-sided variance bound

Chebyshev's inequality and the bound by 1 give, for $a > 0$,

$$\mathbb{P}(|Z - \mathbb{E}[Z]| \geq a) \leq \min\left\{1, \frac{\sigma^2}{a^2}\right\} \leq \frac{2\sigma^2}{\sigma^2 + a^2}.$$

For the last inequality, if $a^2 \leq \sigma^2$, the right-hand side is at least 1; if $a^2 > \sigma^2$, use $\sigma^2 + a^2 \leq 2a^2$. This includes $\sigma^2 = 0$.

(g) General Hoeffding inequality

Set $V = \sum_{k=1}^n (b_k - a_k)^2$ and $\mu = \mathbb{E}[S_n]$. By independence and Hoeffding's lemma (proved in Problem 9),

$$\mathbb{E}[e^{\lambda(S_n - \mu)}] = \prod_{k=1}^n \mathbb{E}[e^{\lambda(X_k - \mathbb{E}[X_k])}] \leq e^{\lambda^2 V / 8}.$$

For $V, t > 0$, exponential Markov inequality followed by $\lambda = 4t/V$ gives

$$\mathbb{P}(S_n - \mu \geq t) \leq e^{-\lambda t + \lambda^2 V / 8} = e^{-2t^2 / V}.$$

Apply the same argument to $-X_k$ and use the union bound:

$$\mathbb{P}(|S_n - \mu| \geq t) \leq 2 \exp\left(-\frac{2t^2}{\sum_{k=1}^n (b_k - a_k)^2}\right).$$

At $t = 0$ the claim is trivial. If $V = 0$, $S_n = \mu$ almost surely, so the tail probability is 0 for $t > 0$ (and 1 for $t = 0$).

Problem 3. Exponential conditional expectations

Use the rate convention $f_{X_i}(x) = \lambda_i e^{-\lambda_i x}$ for $x > 0$. Memorylessness gives $\mathbb{E}[X_i | X_i > t] = t + 1/\lambda_i$, hence

$$\mathbb{E}[X_1 | X_1 > 2026] = 2026 + \frac{1}{\lambda_1}.$$

For truncation below t , integration by parts gives

$$\mathbb{E}[X_1 | X_1 < t] = \frac{\int_0^t x \lambda_1 e^{-\lambda_1 x} dx}{1 - e^{-\lambda_1 t}} = \frac{1}{\lambda_1} - \frac{t}{e^{\lambda_1 t} - 1}.$$

Thus the second answer is $\boxed{1/\lambda_1 - 2026/(e^{2026\lambda_1} - 1)}$. Independence preserves the separate conditional laws under the three threshold events, so

$$\begin{aligned} \mathbb{E}[X_1 + X_2 + X_3 | X_1 > 1997, X_2 > 2014, X_3 > 2026] \\ = \boxed{6037 + \frac{1}{\lambda_1} + \frac{1}{\lambda_2} + \frac{1}{\lambda_3}}. \end{aligned}$$

Problem 4. Bivariate normal construction**(a) Joint normality**

For any $a, b \in \mathbb{R}$,

$$aX + bY = a\mu_X + b\mu_Y + (a\sigma_X + b\rho\sigma_Y)Z_1 + b\sigma_Y\sqrt{1-\rho^2}Z_2.$$

This is a linear combination of independent normal variables, hence normal (possibly constant). Thus (X, Y) is jointly Gaussian.

(b) Correlation

Independence of Z_1, Z_2 gives

$$\text{Var}(X) = \sigma_X^2, \quad \text{Var}(Y) = \sigma_Y^2, \quad \text{Cov}(X, Y) = \rho\sigma_X\sigma_Y, \quad \boxed{\text{Corr}(X, Y) = \rho}.$$

(c) Joint density

Put $u = (x - \mu_X)/\sigma_X$ and $v = (y - \mu_Y)/\sigma_Y$. The inverse map is $z_1 = u$, $z_2 = (v - \rho u)/\sqrt{1-\rho^2}$, with absolute Jacobian $1/(\sigma_X\sigma_Y\sqrt{1-\rho^2})$. Transforming the independent standard-normal density therefore gives, for all $x, y \in \mathbb{R}$,

$$\boxed{f_{X,Y}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left[-\frac{u^2 - 2\rho uv + v^2}{2(1-\rho^2)}\right]}.$$

Problem 5. A three-state Markov chain

With state order $(1, 2, 3)$ and $p_{ij} = \mathbb{P}(X_{n+1} = j | X_n = i)$, the diagram gives

$$P = \begin{pmatrix} 1/4 & 0 & 3/4 \\ 1/2 & 0 & 1/2 \\ 1/2 & 1/4 & 1/4 \end{pmatrix}.$$

(a) One-step probabilities

Time homogeneity gives $\mathbb{P}(X_5 = 3 | X_4 = 2) = p_{23} = \boxed{1/2}$ and $\mathbb{P}(X_4 = 1 | X_3 = 2) = p_{21} = \boxed{1/2}$.

(b) Stationary distribution

Solving $\pi P = \pi$ gives $\pi_2 = \pi_3/4$ and $3\pi_1/4 = \pi_2/2 + \pi_3/2$, hence $\pi_1 = 5\pi_3/6$. Normalization yields

$$\pi = (2/5, 3/25, 12/25).$$

(c) Periods

The positive-probability returns $2 \rightarrow 3 \rightarrow 2$ and $2 \rightarrow 3 \rightarrow 3 \rightarrow 2$ have lengths 2 and 3. Thus $d(2) = \gcd(2, 3) = 1$. All states communicate, so the entire chain has period 1.

(d) Path probability

By the Markov property,

$$\mathbb{P}(X_0 = 3, X_1 = 2, X_2 = 1) = \frac{1}{5} p_{32} p_{21} = \frac{1}{5} \cdot \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{40}.$$

(e) Two-step conditional moments

The first row of P^2 is $(7, 3, 6)/16$. Therefore

$$\mathbb{E}[X_5 \mid X_3 = 1] = \frac{7 + 2 \cdot 3 + 3 \cdot 6}{16} = \frac{31}{16}, \quad \mathbb{E}[X_5^2 \mid X_3 = 1] = \frac{7 + 4 \cdot 3 + 9 \cdot 6}{16} = \frac{73}{16},$$

$$\text{Var}(X_5 \mid X_3 = 1) = \frac{73}{16} - \left(\frac{31}{16}\right)^2 = \frac{207}{256}.$$

Problem 6. Estimating a coin's head probability

Assume $n \geq 1$ and $0 \leq k \leq n$. The likelihood is $L(p) = \binom{n}{k} p^k (1-p)^{n-k}$ on $[0, 1]$.

(a) MLE

For $0 < k < n$, $\ell'(p) = k/p - (n-k)/(1-p) = 0$ gives $p = k/n$; $\ell''(p) < 0$ verifies maximality. If $k = 0$ or n , the maximum is at 0 or 1. Thus $\hat{p}_{\text{MLE}} = k/n$.

(b) MAP

Since $\text{Beta}(1, 1) = \text{Unif}(0, 1)$,

$$f(p \mid X = k) \propto p^k (1-p)^{n-k}, \quad p \mid X = k \sim \text{Beta}(k+1, n-k+1).$$

The uniform prior leaves the likelihood maximizer unchanged, so $\hat{p}_{\text{MAP}} = k/n$, including the endpoint modes on $[0, 1]$.

(c) MMSE

The posterior mean gives $\hat{p}_{\text{MMSE}} = \mathbb{E}[p \mid X = k] = (k+1)/(n+2)$.

(d) LLSE

Under the prior-predictive joint distribution,

$$\mathbb{E}[p] = \frac{1}{2}, \quad \text{Var}(p) = \frac{1}{12}, \quad \mathbb{E}[X] = \mathbb{E}[np] = \frac{n}{2}, \quad \text{Cov}(X, p) = \text{Cov}(\mathbb{E}[X \mid p], p) = \frac{n}{12}.$$

The total-variance formula gives

$$\text{Var}(X) = \mathbb{E}[np(1-p)] + \text{Var}(np) = \frac{n}{6} + \frac{n^2}{12} = \frac{n(n+2)}{12}.$$

Consequently,

$$\hat{p}_{\text{LLSE}} = \frac{1}{2} + \frac{n/12}{n(n+2)/12} \left(k - \frac{n}{2} \right) = \frac{k+1}{n+2}.$$

Problem 7. Gaussian MMSE equals LLSE: two methods

Let $\mu_X = \mathbb{E}[X]$, $\mu_Y = \mathbb{E}[Y]$, $\sigma_X^2 = \text{Var}(X) > 0$, and $\beta = \text{Cov}(X, Y)/\sigma_X^2$.

Method 1: an independent Gaussian residual

Define $Z = Y - \mu_Y - \beta(X - \mu_X)$. Then $\mathbb{E}[Z] = 0$ and

$$\text{Cov}(X, Z) = \text{Cov}(X, Y) - \beta \text{Var}(X) = 0.$$

Because (X, Z) is jointly Gaussian, zero covariance implies independence. Thus $\mathbb{E}[Z | X] = 0$, and

$$\mathbb{E}[Y | X] = \mu_Y + \beta(X - \mu_X) = L[Y | X].$$

The last equality also follows from Problem 2(d).

Method 2: completing the square in the conditional density

For $\sigma_Y > 0$ and $|\rho| < 1$, standardize to $U = (X - \mu_X)/\sigma_X$ and $V = (Y - \mu_Y)/\sigma_Y$. Using $u^2 - 2\rho uv + v^2 = (v - \rho u)^2 + (1 - \rho^2)u^2$,

$$f_{V|U}(v | u) = \frac{f_{U,V}(u, v)}{f_U(u)} = \frac{\exp[-(v - \rho u)^2/(2(1 - \rho^2))]}{\sqrt{2\pi(1 - \rho^2)}}.$$

Hence $\mathbb{E}[V | U = u] = \rho u$, and transforming back gives

$$\mathbb{E}[Y | X = x] = \mu_Y + \frac{\rho\sigma_Y}{\sigma_X}(x - \mu_X) = \mu_Y + \beta(x - \mu_X) = L[Y | X = x].$$

For completeness, if $\sigma_Y = 0$ then $Y = \mu_Y$ almost surely. If $|\rho| = 1$, $\text{Var}(V - \rho U) = 0$, so $V = \rho U$ almost surely and the same answer follows. If $\text{Var}(X) = 0$, the ratio is undefined, but both estimates reduce to $\mathbb{E}[Y]$.

Problem 8. Expected return of the uniform-sum program

Let U_i be independent $\text{Unif}(0, 1)$ increments, $S_n = \sum_{i=1}^n U_i$, and $N = \min\{n \geq 1 : S_n > 1\}$. The program returns S_N , not N . The volume of the simplex $\{u_i \geq 0 : \sum_{i=1}^n u_i \leq 1\}$ is $1/n!$, so

$$\mathbb{P}(N > n) = \mathbb{P}(S_n \leq 1) = \frac{1}{n!}, \quad \mathbb{E}[N] = \sum_{n=0}^{\infty} \mathbb{P}(N > n) = e.$$

The event $\{N \geq i\} = \{S_{i-1} \leq 1\}$ depends only on earlier increments and is independent of U_i . By Tonelli's theorem,

$$\mathbb{E}[S_N] = \sum_{i=1}^{\infty} \mathbb{E}[U_i \mathbf{1}_{\{N \geq i\}}] = \frac{1}{2} \sum_{i=1}^{\infty} \mathbb{P}(N \geq i) = \boxed{\frac{e}{2} \approx 1.35914}.$$

Problem 9. Hoeffding's lemma

Suppose $\mathbb{E}[X] = 0$ and $a \leq X \leq b$. If $a = b$, then $X = 0$ almost surely and the claim is immediate. Otherwise $a \leq 0 \leq b$ and convexity gives

$$e^{\lambda x} \leq \frac{b-x}{b-a} e^{\lambda a} + \frac{x-a}{b-a} e^{\lambda b}, \quad a \leq x \leq b.$$

Taking expectations and putting $p = -a/(b-a) \in [0, 1]$ and $t = \lambda(b-a) > 0$,

$$\mathbb{E}[e^{\lambda X}] \leq (1-p)e^{-pt} + pe^{(1-p)t} = e^{h(t)}, \quad h(t) = -pt + \log(1-p+pe^t).$$

We have $h(0) = h'(0) = 0$ and

$$h''(t) = \frac{p(1-p)e^t}{(1-p+pe^t)^2} = q(t)(1-q(t)) \leq \frac{1}{4}, \quad q(t) = \frac{pe^t}{1-p+pe^t} \in [0, 1].$$

Integrating twice yields $h(t) = \int_0^t (t-s)h''(s) ds \leq t^2/8$. Therefore

$$\mathbb{E}[e^{\lambda X}] \leq \exp\left(\frac{\lambda^2(b-a)^2}{8}\right).$$

Problem 10. Utility, fairness, and reinforcement-learning rewards

Throughout, $\mathcal{F} = \{(x_1, x_2) \in \mathbb{R}_{\geq 0}^2 : x_1 + 2x_2 \leq 12\}$; logarithmic objectives require $x_1, x_2 > 0$.

(a) What a utility function represents

A utility function represents deterministic preferences when

$$x \succeq y \iff u(x) \geq u(y).$$

Any strictly increasing transformation preserves this ranking. Thus x and $\log x$ both mean “more service is better,” but their sums need not have the same maximizer: on $\mathcal{F}$, they yield $(12, 0)$ and $(6, 3)$, respectively (derived in part (c)). Individual rankings do not specify utility differences or comparisons between users; utility scales and weights supply these extra interpersonal trade-offs.

(b) Concavity, diminishing returns, and transfers

The conditions $u'(x) > 0$ and $u''(x) < 0$ mean that service increases utility, but its marginal benefit decreases. Figure 1 shows a representative curve and the decreasing tangent slopes.

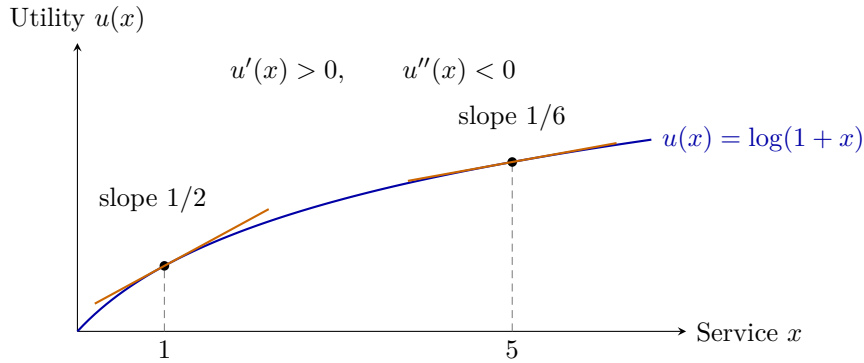


Figure 1: Increasing, strictly concave utility: the same extra service has a smaller marginal benefit at a higher service level.

For $0 < a < b$ and $0 < \delta < (b-a)/2$, define $H(\delta) = u(a+\delta) + u(b-\delta)$. Since $a+\delta < b-\delta$ and u' is strictly decreasing,

$$H'(\delta) = u'(a+\delta) - u'(b-\delta) > 0 \implies u(a+\delta) + u(b-\delta) > u(a) + u(b).$$

This transfer preserves total *service*, not necessarily total *resource use*. For example, moving δ service from user 1 to user 2 increases $x_1 + 2x_2$ by δ . At an interior optimum with equal weights, $u'(x_1) = u'(x_2)/2$: marginal utility per unit of resource, rather than service, is equalized. Thus strict concavity need not imply equal rates; log utility gives $(6, 3)$.

(c) Optimal allocations and proportional fairness

- **Total service:** $x_1 + x_2 \leq x_1 + 2x_2 \leq 12$, with equality only at $(12, 0)$.

- **Log utility:** monotonicity makes the constraint bind. For $f(x_2) = \log(12 - 2x_2) + \log x_2$, the equation $f'(x_2) = -2/(12 - 2x_2) + 1/x_2 = 0$ gives $x_2 = 3$, hence $(6, 3)$. Strict concavity ensures the unique maximum.
- **Minimum service:** if $t = \min\{x_1, x_2\}$, then $3t \leq x_1 + 2x_2 \leq 12$. The bound $t = 4$ is attained at $(4, 4)$.

Objective	Allocation	Total service	Minimum service
Total service	(12, 0)	12	0
Log utility	(6, 3)	9	3
Minimum service	(4, 4)	8	4

Log utility implements *proportional fairness*, neither maximum throughput nor equal service. Indeed, for a maximizer $\mathbf{x}^*$ over a convex $\mathcal{C} \subseteq \mathbb{R}_{>0}^n$, every segment $\mathbf{x}^* + t(\mathbf{y} - \mathbf{x}^*)$, $0 \leq t \leq 1$, is feasible. Optimality gives

$$\left. \frac{d}{dt} \sum_i \log(x_i^* + t(y_i - x_i^*)) \right|_{t=0^+} = \sum_i \frac{y_i - x_i^*}{x_i^*} \leq 0.$$

Thus no feasible change has a positive sum of users' proportional service changes: percentage gains cannot outweigh percentage losses relative to $\mathbf{x}^*$.

For $2 \log x_1 + \log x_2$, the binding constraint and first-order condition give

$$-\frac{4}{12 - 2x_2} + \frac{1}{x_2} = 0 \implies (x_1^*, x_2^*) = (8, 2).$$

The coefficient 2 gives user 1 twice the utility weight, so the same proportional increase counts twice as much. It implements *weighted proportional fairness*; it does not require twice the service or literally represent two users.

(d) Concavity under uncertainty

For the certain payoff A and lottery B,

$$\mathbb{E}[X_A] = \mathbb{E}[X_B] = 5, \quad \mathbb{E}[\log X_A] = \log 5 > \frac{1}{2} \log 1 + \frac{1}{2} \log 9 = \log 3.$$

Expected-payoff maximization is indifferent; expected-log-utility maximization strictly prefers A. The certainty equivalent of B is $c = \exp(\mathbb{E}[\log X_B]) = 3$. Generally, Jensen's inequality gives $\mathbb{E}[u(X)] \leq u(\mathbb{E}[X])$ for concave u : the certain mean is weakly preferred to the lottery (strictly so for strictly concave u and a nonconstant lottery, when expectations exist). This is *risk aversion across outcomes for one user*; part (b) concerns *fairness across different users*.

(e) The von Neumann–Morgenstern theorem

For lotteries P, Q, R over a finite outcome set $\{z_1, \dots, z_m\}$, the four standard preference axioms are:

1. **Completeness:** $P \succeq Q$ or $Q \succeq P$ for every pair.
2. **Transitivity:** $P \succeq Q$ and $Q \succeq R$ imply $P \succeq R$.
3. **Continuity:** if $P \succ Q \succ R$, some $\alpha \in (0, 1)$ satisfies $Q \sim \alpha P + (1 - \alpha)R$.
4. **Independence:** for every $\alpha \in (0, 1)$, $P \succeq Q$ if and only if $\alpha P + (1 - \alpha)R \succeq \alpha Q + (1 - \alpha)R$.

These axioms are equivalent to an expected-utility representation:

$$P \succeq Q \iff \sum_j P(z_j)u(z_j) \geq \sum_j Q(z_j)u(z_j).$$

The utility index is unique up to positive affine transformations $v = au + b$, $a > 0$. The theorem requires neither concavity nor logarithmic utility and does not determine interpersonal fairness. An arbitrary increasing transformation preserves deterministic rankings but may change lottery preferences, because it need not commute with expectation: part (d) has $A \sim B$ under $u(x) = x$ but $A \succ B$ under $v(x) = \log x$.

(f) From utility to episodic RL rewards

The *reward hypothesis* proposes that a goal can be expressed as maximizing the expected cumulative sum of a scalar reward. In this undiscounted, fixed-horizon setting, $J(\pi) = \mathbb{E}_\pi[\sum_{t=0}^{T-1} R_{t+1}]$. Like VNM, this represents preferences by an expectation of a scalar quantity. However, VNM alone supplies neither temporal additivity nor a stationary Markov reward on a prescribed state space.

The two fairness objectives are

$$J_{\text{instant}}(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{T-1} (\log X_1(t) + \log X_2(t)) \right],$$

$$J_{\text{average}}(\pi) = \mathbb{E}_\pi [\log \bar{X}_1 + \log \bar{X}_2], \quad \bar{X}_i = \frac{1}{T} \sum_{t=0}^{T-1} X_i(t).$$

For $T = 2$, both schedules have average allocation $(6, 3)$:

Schedule	Allocations	J_{instant}	J_{average}
π_A	$(6, 3), (6, 3)$	$2 \log 18 = \log 324$	$\log 18$
π_B	$(10, 1), (2, 5)$	$2 \log 10 = \log 100$	$\log 18$

Thus the objectives are not equivalent: the first strictly prefers A, while the second is indifferent. Even dividing J_{instant} by T does not remove this difference; concavity penalizes temporal fluctuations under instantaneous utility.

Instantaneous utility uses $R_{t+1} = \log X_1(t) + \log X_2(t)$. For average-service utility, let $C_i(t) = \sum_{s=0}^{t-1} X_i(s)$, with $C_i(0) = 0$, and use the terminal reward

$$R_{t+1} = \begin{cases} 0, & 0 \leq t < T-1, \\ \log \frac{C_1(t) + X_1(t)}{T} + \log \frac{C_2(t) + X_2(t)}{T}, & t = T-1. \end{cases}$$

Pathwise, $\sum_{t=0}^{T-1} R_{t+1} = \log \bar{X}_1 + \log \bar{X}_2$, so its expectation is exactly J_{average} . Augment a Markov environment state to $\tilde{S}_t = (S_t, C_1(t), C_2(t), t)$, updating $C_i(t+1) = C_i(t) + X_i(t)$. The accumulated service and clock make this a Markov reward with no additional external time dependence; complete histories are unnecessary.

(g) VNM and the reward hypothesis: connection and limits

Treat a complete trajectory τ as an outcome and a policy as inducing a lottery over trajectories. When the relevant expected-utility representation exists, preferences take the form $\mathbb{E}_\pi[U(\tau)]$; RL realizes this form when $U(\tau) = \sum_t R_{t+1}$. For example, part (f)'s terminal reward realizes utility of average service exactly. VNM is a conditional representation theorem, not a proof that every goal admits an additive stationary Markov reward on the original state space. Temporal structure, sufficient state information, and fairness weights are additional modeling choices. Moreover, applying the finite-outcome theorem to continuous trajectory spaces requires appropriate additional assumptions.